FINAL YEAR PROJECT (BSC CSE): REVIEW FINDINGS AND PROTOTYPE SCOPING PROPOSAL

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1. BACKGROUND AND PURPOSE OF THIS DOCUMENT

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Following our last meeting, we reviewed past student projects in the UMaT Library archives from 2015 to 2024, since the 2025 and 2026 reports have not been catalogued by the library yet. We searched across the Computer Science and Engineering (CSE), Information Systems, Cybersecurity, and Mining Engineering departments to see if any previous work could guide our two proposed topics.

This document summarizes our findings, explains the key technical and mining terms, and outlines a clear, realistic scope for the prototypes we plan to build.

2. TOPIC 1: Development of a Smart Family-Safe Media Player Using Computer Vision and Audio Signal Processing for Automated Explicit Content Filtering

2.1 PROPOSED PROJECT TITLE

Development of a Smart Family-Safe Media Player Using Computer Vision and Audio Signal Processing for Automated Explicit Content Filtering

2.2 UMAT LIBRARY ARCHIVE FINDINGS (2015 TO 2024)

Search Scope: CSE, IS, Cybersecurity, and Mining Engineering project catalogues from 2015 to 2024.

Finding: When we checked the available 2015 to 2024 project lists across these departments, we did not come across any past project that built an AI-powered video player for automatic visual and audio filtering of explicit content. Once the library finishes cataloguing the 2025 and 2026 project reports, we can check those batches too just to be sure no recent group worked on something similar.

Project Significance: This gives our group a very fresh software and AI project direction at UMaT while solving a real everyday problem for parents, families, and schools.

2.3 Key Technical Terms and Jargon Explained

Multimodal Filtering: Analyzing more than one type of media signal at the same time. In our case, it means combining Computer Vision (checking video frames) and Audio Signal Processing (checking sound and speech).

Pretrained Image Classification Model (Yahoo Open NSFW): A deep learning Convolutional Neural Network (CNN) built by Yahoo that takes an image or video frame and gives a probability score between **“0.0”** (general/safe) and **“1.0”** (explicit/NSFW). Using an existing pretrained model means we do not have to gather explicit images ourselves or train a heavy model from scratch.

Frame Sampling: Instead of checking all 30 or 60 frames in every single second of a movie (which would slow down a normal laptop), the player picks **1 or 2** frames per second to detect explicit scenes smoothly.

Audio Event Classification and Speech-to-Text (STT): Using lightweight, open-source AI models (like Google's **YAMNet** for non-verbal sounds or **Whisper/Vosk** for speech transcription) to catch profanity or explicit audio sounds and mute the sound automatically.

Video Metadata and Local Content Database: Basic details stored inside a video file (like the file title, duration, and file hash) that our player can read and compare against a local database of known movies.

2.4 PROPOSED MANAGEABLE PROTOTYPE SCOPE

To keep the project practical, unique, and realistic for a student prototype (tested on short 1 to 3 minute video clips), our proposed system will include:

1. Open Desktop Video Player: A clean desktop video player that opens and plays supported video files **(“.mp4”, “.mkv”, “.avi”)** right away, just like VLC or any normal media player, without forcing regular users to log in before watching a video.

2. AI Visual Censorship Engine: Integration of the **Yahoo Open NSFW image-classification model** to automatically black out, blur, or skip video scenes when explicit visual content is detected.

3. Lightweight Audio and Profanity Filtering: Integration of an open-source pretrained audio or speech model to mute profanity or explicit non-verbal sounds during playback.

4. Optional Parental Account Login and PIN Lock System: While the video player stays open for anyone to use normally, a parent setting up the app for their child or ward can open the "**Parental Control Setup**" menu. There, the parent can create a simple local account login and set a **4-digit PIN**. Once locked, the child cannot turn off the AI filter, lower the sensitivity settings (Low, Medium, Strict), or delete the filtering history without entering the **parent's login** or **PIN**.

5. **Automated Filtering History Log:** A timestamped summary saved inside the Parental Control section so parents can see what was filtered while their ward was watching a movie. For example:

`Movie: Titanic.mp4`

`00:42 - Explicit visual content detected (Blurred)`

`03:17 - Profanity/Explicit audio detected (Muted)`

`07:51 - Explicit visual content detected (Blurred)`

`Total filtered events: 3`

6. **Local Movie Metadata Database:** Alongside the AI model, we plan to include a simple local database (using SQLite or JSON) filled with movie details that we can gather from public online platforms like the **IMDb Parents Guide**, **TMDB API**, **Kids-In-Mind**, or **DoesTheDogDie.**

What the Database Will Hold: **Movie Title, Release Year, Age Rating** (**like PG-13 or 18+**), **Content Advisory Tags** (such as **Nudity**: Severe, Language: **Moderate**), and known explicit scene timestamps where available.

How It Works: When someone loads a movie, the player checks the video's title or metadata against our local database first. If the movie is in the database, it shows a quick content warning and loads known timestamps right away. If the movie is not in the database, the AI model scans the video frames as it plays. We can also propose a monthly database update feature so new movies can be added over time.

2.5 PROPOSED FUTURE SCOPE

Fast Offline AI Auto-Subtitling: Adding a fast speech-to-text model so the player can generate English subtitles automatically for movies that do not have `.srt` subtitle files.

Real-Time Live Streaming, Browser Extension, and Smart TV OS Support: Extending the system into a browser extension and building it into Smart TV Operating Systems **(Android TV / WebOS)** and higher-end laptops that have enough processing power to run live video and audio AI filtering at the same time without lagging.

More Filtering Categories: Expanding the AI detection to cover extreme violence, blood/gore, and drug use.

2.6 VERIFIED REFERENCES FOR TOPIC 1

1. Mahadeokar, J. and Pesavento, G. (2016), "Open Sourcing a Deep Learning Solution for Detecting NSFW Images", Yahoo Engineering Technical Blog, Yahoo Inc., retrieved from `[https://yahooeng.tumblr.com/post/151148689421/open-sourcing-a-deep-learning-solution-for](https://yahooeng.tumblr.com/post/151148689421/open-sourcing-a-deep-learning-solution-for)`. (Primary reference for the Yahoo Open NSFW Convolutional Neural Network model).

2. Hershey, S., Chaudhuri, S., Ellis, D. P. W., Gemmeke, J. F., Jansen, A., Moore, R. C., Plakal, M., Platt, D., Saurous, R. A., Seybold, B., Slaney, M., Weiss, R. J. and Wilson, K. (2017), "CNN Architectures for Large-Scale Audio Classification", Proceedings of the IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP), New Orleans, LA, USA, pp. 131-135. (Primary reference for pretrained audio event classification models such as YAMNet).

3. Gemmeke, J. F., Ellis, D. P. W., Freedman, D., Jansen, A., Lawrence, W., Moore, R. C., Plakal, M. and Ritter, M. (2017), "Audio Set: An Ontology and Human-Labeled Dataset for Audio Events", Proceedings of the IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP), New Orleans, LA, USA, pp. 776-780. (Primary reference for the dataset used by pretrained audio event classifiers).

4. Radford, A., Kim, J. W., Xu, T., Brockman, G., McLeavey, C. and Sutskever, I. (2023), "Robust Speech Recognition via Large-Scale Weak Supervision", Proceedings of the 40th International Conference on Machine Learning (ICML), PMLR, Vol. 202, pp. 28492-28518. (Primary reference for automated speech transcription and profanity timestamping).

3. TOPIC 2: Design and Implementation of an IoT-Based Multi-Sensor Tailings Storage Facility (TSF) Freeboard and Overflow Pre-Alarm System

3.1 PROPOSED PROJECT TITLE:

Design and Implementation of an IoT-Based Multi-Sensor Tailings Storage Facility (TSF) Freeboard and Overflow Pre-Alarm System

3.2 UMAT LIBRARY ARCHIVE FINDINGS (2015 TO 2024) AND HOW THEY SUPPORT OUR WORK

We found two (2) past UMaT student projects that directly relate to this topic, one from **Computer Science and Engineering (2017)** and one from **Mining Engineering (2024)**, along with **2025 public TSF data from AngloGold Ashanti Iduapriem Mine.** Together, they give us a solid foundation to justify this project to our supervisor:

1. Past CSE Project: **Addo, C. (2017), "Remote Control and Monitoring of Water Reservoir System Using a Mobile App and a Microcontroller"* (Supervised by Mr. Richard Kofi Annan).**

What He Built: A **tabletop prototype** using **two plastic containers**, an **Arduino Uno R3**, **bare aluminum strips** and **CAT 5 wires** placed directly inside the water as a **level sensor**, an **HC-05 Bluetooth module** (with a short range of about 10 meters), a **12V water pump**, and an Android app called "**WaterRez"**.

How It Guides Our Proposal: In Chapter 5 (Section 5.2, Page 34), **Addo** specifically recommends that future UMaT students use a **Wi-Fi module** to give a wider range of remote access and **an Ultrasonic (contactless) sensor** to read liquid levels so that metal sensors do not have to be cleaned regularly or suffer from corrosion. In our proposed project, we are aiming to implement these exact recommendations from **Addo (2017)** and apply them to a mining safety problem.

2. Past Mining Engineering Project: **Ababio, O. T., Abdul Hamid, I. and Akrong, A. (2024), "Review of Tailings Storage Facility Construction and Management - A Case Study" (Supervised by Dr. Richard Gyebuni).**

What They Studied: A detailed engineering and safety review of the **Green Fields Tailings Storage Facility (GTSF)** and Beposo **TSF at AngloGold Ashanti Iduapriem Limited (AAIL)** right here in Tarkwa.

JUSTIFICATION OF PROPOSED PROJECT:

Real Tarkwa Data on Freeboard and Overtopping (Section 4.6, Page 38): In simple terms, a Tailings Storage Facility (TSF) is a huge earth-walled dam where a mine pumps its muddy, chemical-filled liquid waste (tailings) after extracting gold. Freeboard is the empty vertical safety space between the top surface of that liquid waste and the top edge (crest) of the dam wall. If heavy rain or pump failure causes the liquid to rise and fill up that safety gap, the liquid pours over the top of the earth wall.

That overflow is called Overtopping, and because the dam wall is built from compacted soil and rock (like saprolite and laterite) rather than concrete, overflowing liquid washes the wall away and can cause the whole

dam to collapse. Ababio et al. show that at Iduapriem Mine's GTSF, the daily freeboard is normally kept between 2.5 m and 3.5 m, with a strict minimum safety threshold of 1.5 m. If the freeboard gap drops below 1.5 m, there is a serious risk of overtopping and spillage.

Global Failure Statistics (Section 3.1, Page 9): Citing Stark et al. (2022), the report shows a global chart where Overtopping (OT) is responsible for 54 major tailings dam failures around the world, making it one of the highest known causes of dam disasters.

RECOMMENDATIONS FROM LITERATURE (CHAPTER 5, PAGES 44 TO 46):

In their conclusions (Section 5.1, Page 45), Ababio et al. point out that overtopping or overflow, slope instability, and earthquakes are the main failure modes that could threaten the Iduapriem TSF, especially because the tailings have high water content and high pore-water pressure. To improve safety, they list innovative technologies that mines should adopt, specifically citing Dong et al. (2017), who proposed a pre-alarm system that uses the Internet of Things (IoT), Cloud Computing, AI, and real-time sensor data to track tailings dam conditions and warn operators before an overflow or failure happens. Then, in their recommendations (Section 5.2, Page 46), they advise the mine to build Best Available Technology (BAT) into their safety practices and close any monitoring gaps.

The 2024 Mining students identified the exact overflow risk at Iduapriem Mine and pointed to an IoT and Cloud pre-alarm system as the solution, but since they are in Mining Engineering, their work was a case study review rather than a hardware build. As CSE students, our group can use their paper to justify designing and building the actual IoT pre-alarm prototype they recommended.

3.3 SUMMARY OF KEY TERMS AND TECHNICAL JARGONS

Tailings and TSF (Tailings Storage Facility): The muddy liquid waste left over after mineral processing, stored inside a large earthen dam **(for example, AngloGold Ashanti's Greenfields TSF in Tarkwa, located at Lat. 5.233688°, Long. -2.066815°).**

Freeboard: The vertical safety gap between the liquid level in the dam and the top edge of the dam wall (1.5 m minimum threshold at Iduapriem Mine).

Overtopping: When liquid rises above the dam crest and spills over the wall, eroding the earth structure (54 major cases recorded worldwide).

Decant System and Pore Pressure: The pumping system used to remove excess surface water from the dam. If too much water stays in the dam and soaks into the soil wall, internal water pressure (**pore pressure**) builds up and weakens the embankment.

3.4 PROPOSED MANAGEABLE PROTOTYPE SCOPE

Just like Addo (2017) used two plastic containers for his reservoir prototype **(costing GH₵ 241 in 2017)**, we plan to build a clean, low-cost tabletop scale model of a tailings dam pre-alarm system without overcomplicating the hardware:

1. **Microcontroller and Dual Communication:** An **ESP32** microcontroller (using its built-in Wi-Fi to send live sensor readings to a Web Dashboard) paired with a **GSM Module** (such as SIM800L) so the system can send **emergency SMS** or phone call alerts to mining operators if the internet goes down.

2. **Simple Multi-Sensor Setup (3 Affordable Sensors):**

(i) **Waterproof Ultrasonic Sensor (JSN-SR04T):** Mounted at the top edge of our model container to measure the **Freeboard distance** without touching the muddy water (avoiding metal corrosion and scaling Iduapriem's 1.5 m safety threshold down to our container size).

(ii) **Soil Moisture Sensor:** Placed in the soil wall of our model to detect water seepage and saturation in the dam embankment.

(iii) **Rain Drop / Rain Gauge Sensor:** Placed at the top of the setup to detect rainfall, giving an early warning of extra water entering the dam.

3. Local Alarm and Mini Decant Pump (Optional Relay): LED lights (Green for Safe Freeboard, Yellow for Warning, Red for Critical Overflow Risk), a **buzzer**, and a **small 12V** submersible pump connected through a relay to pump excess water out into a second container when the critical level is reached (representing Iduapriem's decant system).

4. Web-Based Monitoring Dashboard: A clean web application that can be opened on any laptop or phone browser anywhere there is internet, showing live freeboard levels, wall moisture status, rain alerts, and past alert logs.

3.5 PROPOSED FUTURE SCOPE

LoRa (Long Range) Wireless Network: Proposing LoRa radio modules for long-distance wireless data transmission across remote mining areas where cellular networks and Wi-Fi do not reach.

Offline Mobile App with Local LoRa Connection: Proposing a dedicated mobile app for field workers that connects to a local LoRa receiver inside the mine site, allowing the app to receive and process sensor readings offline without needing internet access.

3.6 VERIFIED REFERENCES FOR TOPIC 2

1. Addo, C. (2017), "Remote Control and Monitoring of Water Reservoir System Using a Mobile App and a Microcontroller", Unpublished BSc Project Report, Department of Computer Science and Engineering, University of Mines and Technology (UMaT), Tarkwa, Ghana, 39 pp. (Specifically citing Section 3.3, pp. 17-22; Section 4.2, p. 25; Section 4.6, p. 32; and Section 5.2, p. 34).

2. Ababio, O. T., Abdul Hamid, I. and Akrong, A. (2024), "Review of Tailings Storage Facility Construction and Management - A Case Study", Unpublished BSc Project Report, Department of Mining Engineering, University of Mines and Technology (UMaT), Tarkwa, Ghana, 50 pp. (Specifically citing Section 2.6.1, p. 8; Section 3.1, p. 9; Section 4.6, p. 38; Section 4.8, pp. 40-41; and Chapter 5, pp. 44-46).

3. Stark, T. D., Moya, L. and Lin, J. (2022), "Rates and Causes of Tailings Dam Failures"*, Advances in Civil Engineering, Vol. 2022, No. 1, Article ID 6619412, 21 pp. (Cited in Ababio et al., 2024, Fig. 3.1, p. 9, documenting 54 global tailings dam failures caused by Overtopping).

4. Dong, L., Shu, W., Sun, D., Li, X. and Zhang, L. (2017), "Pre-Alarm System Based on Real-Time Monitoring and Numerical Simulation Using Internet of Things and Cloud Computing for Tailings Dam in Mines", IEEE Access, Vol. 5, pp. 21080-21089. (Cited in Ababio et al., 2024, Section 5.1, p. 45, proposing IoT, Cloud Computing, AI, and real-time sensor pre-alarm systems for tailings dams).

5. AngloGold Ashanti (2025), "TSF Disclosures: Greenfields TSF, Ghana (At June 2025)", Official Sustainability and Environment Disclosure, retrieved from `[https://www.anglogoldashanti.com](https://www.anglogoldashanti.com)` (Coordinates: Lat. 5.233688°, Long. -2.066815°).